

KOCHUBEY, Russian FederationWMO: **370850**Lat: **44.400N**Lon: **46.550E**Elev: **-21**StdP: **101.58**Time Zone: **3.00 (E03)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.1	-12.0	-18.4	0.7	-15.8	-14.4	1.1	-11.3	10.7	3.0	9.4	2.2	2.2	300	0.487

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	35.3	21.5	33.8	21.2	32.5	21.0	24.0	30.8	23.2	29.9	22.5	29.1	4.5	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.1	16.7	26.9	21.2	15.8	26.3	20.3	15.0	25.8	72.0	31.1	68.8	29.6	66.1	29.2	27.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	10.3	8.9	7.6	DB	-19.1	38.2	5.2	1.9	-22.8	39.5	-25.9	40.6	-28.8	41.6	-32.7	43.0
(5)				WB	-19.3	26.1	5.2	0.7	-23.1	26.6	-26.1	26.9	-29.0	27.3	-32.8	27.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.6	-0.5	0.0	5.0	11.4	18.4	23.7	26.2	25.6	20.0	13.0	6.1	1.8
	DBStd	10.43	5.23	5.44	3.62	3.79	3.65	2.80	2.46	2.94	3.51	4.54	4.69	4.76
	HDD10.0	1194	326	279	158	26	1	0	0	0	0	22	129	253
	HDD18.3	2832	584	512	412	209	43	1	0	0	22	170	366	511
	CDD10.0	2160	0	0	4	68	263	411	501	483	301	116	13	1
	CDD18.3	757	0	0	0	2	47	163	242	224	73	6	0	0
	CDH23.3	7657	0	0	1	27	404	1576	2702	2429	498	22	0	0
	CDH26.7	3282	0	0	0	3	101	623	1297	1131	124	2	0	0
Wind	WSAvg	3.2	3.0	3.3	3.5	3.3	3.2	3.2	3.1	3.0	3.2	3.0	3.1	3.2
Precipitation	PrecAvg	283	18	14	18	22	33	37	27	23	27	24	20	20
	PrecMax	400	33	44	46	57	104	120	87	76	92	65	52	44
	PrecMin	171	2	4	1	0	3	5	4	2	1	1	1	6
	PrecStd	59	8	9	12	15	19	28	22	21	25	17	14	10
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.3	11.9	19.4	26.7	31.7	35.3	37.8	37.2	33.1	26.0	17.8	12.7
		MCWB	7.1	7.8	11.2	15.3	18.6	21.0	22.2	21.9	22.2	18.5	13.4	10.2
	2%	DB	7.8	9.2	14.9	23.4	29.4	33.5	35.4	35.3	29.6	23.1	15.3	10.3
		MCWB	5.9	6.2	9.7	14.5	18.1	20.6	21.6	21.6	20.2	17.4	12.5	8.6
	5%	DB	6.4	7.3	12.3	20.8	27.3	31.8	33.9	33.8	27.7	21.1	13.8	8.6
		MCWB	5.0	5.3	8.1	13.5	17.5	19.7	21.0	21.7	19.5	16.3	11.6	7.1
	10%	DB	5.2	5.9	10.3	18.4	25.5	30.2	32.6	32.2	26.1	19.4	12.4	7.3
		MCWB	4.2	4.5	7.0	12.4	16.9	19.2	20.9	21.5	18.7	15.6	10.6	6.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.5	8.6	12.1	16.4	20.7	23.3	24.9	25.2	23.8	19.3	14.5	10.7
		MCDB	9.4	11.6	18.3	23.5	27.4	31.4	33.2	31.5	30.0	23.7	16.5	12.0
	2%	WB	6.1	6.6	10.0	15.2	19.5	22.2	23.8	24.3	22.1	18.2	13.0	8.8
		MCDB	7.5	8.5	14.3	22.3	26.3	29.9	31.3	30.6	27.2	22.1	14.9	10.2
	5%	WB	5.2	5.6	8.6	14.0	18.5	21.4	23.0	23.5	20.7	17.2	11.9	7.2
		MCDB	6.2	7.0	11.7	19.7	25.0	28.5	30.0	29.8	25.6	20.3	13.6	8.2
	10%	WB	4.2	4.7	7.4	12.9	17.7	20.6	22.2	22.7	19.8	16.0	10.7	6.2
		MCDB	5.1	5.8	9.8	17.5	23.9	27.3	29.1	29.1	24.5	18.8	12.2	7.2
Mean Daily Temperature Range	5% DB	MDBR	5.4	6.1	7.5	9.8	10.6	10.9	11.0	11.3	9.9	8.4	6.4	5.2
		MCDBR	7.2	8.3	11.2	14.0	13.1	13.1	13.3	11.4	9.9	8.4	6.7	
		MCWBR	5.2	5.6	6.5	6.7	4.4	4.1	3.9	4.2	4.6	5.0	5.7	5.1
	5% WB	MCDBR	6.4	7.1	10.3	12.4	11.8	11.5	11.6	11.9	10.0	9.3	7.6	5.9
		MCWBR	4.9	5.2	6.3	6.4	4.7	4.5	4.4	5.0	4.8	5.1	5.4	4.7
Clear-Sky Solar Irradiance	taub	0.372	0.403	0.434	0.499	0.467	0.466	0.487	0.494	0.465	0.417	0.383	0.383	
	taud	2.184	2.105	2.071	1.941	2.105	2.166	2.120	2.088	2.117	2.201	2.223	2.154	
	Ebn at Noon	708	754	788	767	810	811	788	765	750	733	691	644	
	Edh at Noon	94	122	143	177	155	146	152	151	136	109	90	89	
All-Sky Solar Radiation	RadAvg	1.12	1.86	3.14	4.76	5.94	6.61	6.43	5.79	4.32	2.59	1.39	0.94	
	RadStd	0.19	0.32	0.32	0.50	0.56	0.36	0.39	0.29	0.37	0.25	0.18	0.12	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page